

STEP 1: Clear everything

```
clc;  
clear all;  
close all;
```

Clears command window, variables, and closes all figures.

STEP 2: Read and Display Original Image

```
img = imread('peppers.png');  
figure(1);  
imshow(img);  
title('Original Image');
```

Loads and displays the original image.

STEP 3: Convert to Grayscale

```
if size(img, 3) == 3  
    gray_img = rgb2gray(img);  
else  
    gray_img = img;  
end
```

Converts image to grayscale.

STEP 4: Noise Removal using Gaussian Filter

```
filtered_img = imgaussfilt(gray_img, 2);
```

Applies Gaussian filter to remove noise.

STEP 5: Thresholding using Otsu's Method

```
threshold = graythresh(filtered_img);  
binary_img = imbinarize(filtered_img, threshold);
```

Converts image to binary using Otsu's method.

STEP 6: Morphological Operations

```
se = strel('disk', 5);  
cleaned = imopen(binary_img, se);  
cleaned = imclose(cleaned, se);
```

Cleans the binary image using morphological operations.

STEP 7: Label Connected Regions

```
labeled = bwlabel(cleaned);  
stats = regionprops(labeled, 'Area', 'BoundingBox', 'Centroid');  
Detects and labels objects in the image.
```

STEP 8: Overlay Segmented Objects

```
imshow(img);
hold on;
for i = 1:length(stats)
    bb = stats(i).BoundingBox;
    rectangle('Position', bb, 'EdgeColor', 'r', 'LineWidth', 2);
    plot(stats(i).Centroid(1), stats(i).Centroid(2), 'g+', ...
        'MarkerSize', 10, 'LineWidth', 2);
end
hold off;
```

Draws bounding boxes and centroids on detected objects.